Unified Information-Density Theory (UIDT) v3.9

Collaboration Request
Lattice-QCD Determination of $|\psi_{cc}(0)|^2$ for the Ξ_{cc} Diquark System

via Extended HAL QCD Method (Kelvin-Lee & Ishii 2026)

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Addressed to:	Prof. N. Ishii, K.-W. Kelvin-Lee HAL QCD Collaboration RCNP, Osaka University

Evidence Classification: *Stratum I (LHCb input) + Stratum III (UIDT mapping). Central claim: Evidence Category B, with explicit upgrade path to A^- conditional on Lattice result.*

Abstract. The UIDT Framework (v3.9) predicts the isospin mass splitting of the doubly charmed baryon Ξ_{cc} as $\Delta M_{\mathrm{UIDT}} = 1.504\,\mathrm{MeV}$ via a four-term chromomagnetic decomposition, in agreement with the LHCb measurement $\Delta M_{\mathrm{emp}} = 1.43 \pm 0.76\,\mathrm{MeV}$ (residuum $0.074\,\mathrm{MeV}$, $< 0.10\,\sigma$). The dominant contribution (89.8%) originates from the chromomagnetic term, which depends on the cc-diquark Nambu–Bethe–Salpeter (NBS) wave function at the origin:

$$|\psi_{cc}(0)|^2_{\mathrm{UIDT}} = 0.4719\,\mathrm{fm}^{-3}.$$

Currently computed via Coulomb approximation ($\alpha_s = 0.30$, $M_c = 1500\,\mathrm{MeV}$), this is the sole uncontrolled element of the calculation. We request a first-principles Lattice-QCD determination using the extended HAL QCD method [1], which would upgrade the claim from Evidence B to A^- and enable a decisive test of the UIDT vacuum parameter $\Delta^* = 1.71003\ldots \pm 0.015\,\mathrm{GeV}$. All computations are fully reproducible at 80-digit `mpmath` precision; no float conversion is used anywhere in the pipeline.

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1 Introduction and Motivation

The UIDT Framework introduces a fundamental scalar field $S(x)$ representing vacuum information density, extending standard Yang-Mills dynamics through non-minimal coupling. Its canonical parameters, established by the Banach fixed-point iteration and phenomenological calibration [6], are:

UIDT Immutable Parameter Ledger — Banach 80-digit Precision

Δ^*	$= 1.71003504674221318202077109661162\ldots \pm 0.015\,\text{GeV}$	**[Ev. A]**
γ	$= 16.339$	**[Ev. A-]**
γ_∞	$= 16.3437$	**[Ev. A-]**
$\delta\gamma$	$= 0.0047$	**[Ev. A-]**
v	$= 47.7\,\text{MeV}$	**[Ev. A]**
w_0	$= -0.99$	**[Ev. C]**
E_T	$= 2.44\,\text{MeV}$	**[Ev. C]**

Δ^*: Yang–Mills spectral gap (not a particle mass); full 240-digit value available in verification suite.
γ: phenomenological kinetic vacuum parameter (ratio Δ^*/K_S).
Ledger values are immutable; no automatic modification permitted by UIDT constitution.

A recent internal analysis computed the isospin splitting of Ξ_{cc} at 80-digit `mpmath` precision using a four-term decomposition of the chromomagnetic interaction. The result agrees with the LHCb measurement at the $0.10\,\sigma$ level, but the dominant term rests on a Coulomb approximation for the cc-diquark NBS wave function. Replacing this approximation with a direct Lattice-QCD determination using the extended HAL QCD method of Kelvin-Lee & Ishii [1] would close the last uncontrolled systematic in the UIDT Ξ_{cc} prediction.

2 The Ξ_{cc} Isospin Splitting in UIDT

2.1 Four-Term Decomposition

The isospin mass splitting is written as:

$$\Delta M(\Xi_{cc}) = \delta\chi_{\text{em}} + \delta\chi_{\text{qcd}} + \delta\chi_{\text{cmag}} + \delta\chi_\pi \tag{1}$$

2.2 The Coulomb NBS Approximation

The chromomagnetic term uses the Coulomb wave function at the origin:

$$|\psi(0)|^2_{\text{Coulomb}} = \frac{(\alpha_s M_c/2)^3}{\pi} = \frac{(0.30 \times 1500/2)^3}{\pi} = 3{,}625{,}748\ \text{MeV}^3 = 0.4719\,\text{fm}^{-3} \tag{2}$$

with $\alpha_s = 0.30$ (PDG, $\mu \approx M_c$) and $M_c = 1500\,\text{MeV}$.

Limitation 2.1. Equation (2) treats the cc-diquark system as a hydrogen-like Coulomb bound state. This neglects: (i) non-perturbative string tension beyond $r \sim 0.2\,\text{fm}$, (ii) relativistic corrections $\mathcal{O}(v^2/c^2) \sim 15\%$, (iii) finite-size effects of the diquark. The Lattice-QCD NBS method resolves all three simultaneously.

Term	**Formula**	**Value [MeV]**	**Note**
$\delta\chi_{\text{em}}$	$\frac{\alpha_{\text{em}}}{\pi} \cdot \Delta Q^2 \cdot \frac{\Delta^*}{\gamma}$	0.0810	EM self-energy **[Ev. A]**
$\delta\chi_{\text{qcd}}$	$(m_d - m_u) \cdot \frac{\Delta^*}{\gamma M_{\Xi_{cc}}}$	0.0731	Light-quark mass diff. **[Ev. B]**
$\delta\chi_{\text{cmag}}$	$\frac{4\alpha_s}{3M_c^2} \lvert\psi(0)\rvert^2 \cdot \langle \vec{S}_c \cdot \vec{S}_c \rangle$	1.3500	**Dominant (89.8%) [Ev. B]**
$\delta\chi_\pi$	Pion-loop correction	$< 10^{-5}$	Negligible **[Ev. A]**
ΔM_{UIDT}		**1.504**	
ΔM_{emp}	LHCb 2026	1.43 ± 0.76	Stratum I
Residuum		**0.074**	$< 0.10\,\sigma$

Table 1: Four-term decomposition of $\Delta M(\Xi_{cc})$ in the UIDT Framework. All values computed at 80-digit `mpmath` precision.

3 Technical Specification of the Lattice Calculation

3.1 Target Observable and UIDT Prediction

Definition 3.1 (cc-Diquark NBS Wave Function)**.** The target observable is the equal-time quark–diquark NBS amplitude for the c–(cc) system in Ξ_{cc}:

$$|\psi_{cc}(0)|^2_{\text{lat}} \;\equiv\; |\phi_{cD}(\vec{r}=0)|^2 \tag{3}$$

where $\phi_{cD}(\vec{r})$ is defined as in equations (2.1)–(2.3) of Kelvin-Lee & Ishii [1], adapted to the cc-diquark channel.

Testable Prediction 1 (cc-Diquark NBS at Origin)**.** The UIDT Framework predicts:

$$\boxed{|\psi_{cc}(0)|^2_{\text{UIDT}} = 0.4719\,\text{fm}^{-3}} \tag{4}$$

Falsification window ($\pm 20\%$): $|\psi_{cc}(0)|^2_{\text{lat}} \notin [0.378,\, 0.566]\,\text{fm}^{-3}$.

Falsification Criterion 1 (Kill-Switch Criterion)**.** The prediction is considered falsified if $|\psi_{\text{lat}}|^2 < 0.30\,\text{fm}^{-3}$ or $> 0.70\,\text{fm}^{-3}$ (outside the $2\,\sigma$ band), triggering mandatory review of the Coulomb ansatz in the chromomagnetic term.

3.2 Scenario Table

Lattice Result	**UIDT Status**	**Consequence**
$\lvert\psi\rvert^2 \in [0.378,\, 0.566]\,\text{fm}^{-3}$	Compatible	Claim upgraded to Ev. A$^-$
$\lvert\psi\rvert^2 < 0.378\,\text{fm}^{-3}$	[TENSION ALERT]	Non-Coulombic cc structure indicated
$\lvert\psi\rvert^2 > 0.566\,\text{fm}^{-3}$	[TENSION ALERT]	Additional geometric contribution needed

Table 2: Outcome scenarios for the proposed Lattice-QCD determination.

3.3 Diquark Operator

The cc-diquark in the $\bar{\mathbf{3}}_c$, spin-1 (axial-vector) representation, consistent with Ξ_{cc} ($J^P = 1/2^+$):

$$D_i^{ab}(x) = \epsilon^{abc}\, c^{T,a}(x)\, C\gamma_i\, c^b(x) \tag{5}$$

where $C = \gamma^0\gamma^2$ is the charge conjugation matrix.

3.4 Extraction Protocol

1. **3-point correlator:**
$$\langle D_i(t_f)\, J_{\Xi_{cc}}(t)\, D_i^\dagger(t_i)\rangle$$
following the HAL QCD procedure of Kelvin-Lee & Ishii [1].

2. **NBS amplitude:** Extract $|\phi_{cD}(\vec{r})|^2$ in the range $r \lesssim 0.25\,\text{fm}$.

3. **Short-distance fit:** S-wave Coulomb-type ansatz; Cornell-type (Coulomb + linear) as found in your scalar diquark study [1] is the expected form.

4. **Extrapolation to origin:** Report $|\phi_{cD}(0)|^2$ with statistical and systematic uncertainties from: lattice spacing, chiral extrapolation, finite volume.

3.5 Recommended Gauge Ensemble

$N_f = 2+1$, $a^{-1} \approx 1.6$–$2.0\,\text{GeV}$, clover charm quarks. Your existing CP-PACS/JLQCD configurations are in principle sufficient; PACS-CS at $a^{-1} \approx 2.2\,\text{GeV}$ would reduce discretisation effects. Comparison with CLQCD ensembles at $a = 0.07746\,\text{fm}$ [4] is recommended as a cross-check.

4 Relation to Existing Literature

Reference	Relevance	Connection
Kelvin-Lee & Ishii 2026 [1]	Extended HAL QCD, scalar/axial diquark, Cornell potential	Direct methodological basis
Watanabe & Ishii 2022 [2]	q–D potential, NBS wavefunction extraction	Established pipeline
Padmanath et al. 2015 [3]	Ξ_{cc} operator construction on lattice	Operator basis reference
Yi et al. 2025 [4]	Ξ_{cc} interactions, CLQCD ensembles, $a = 0.07746\,\text{fm}$	Cross-check dataset
Rietz 2026 [6]	UIDT v3.9, all ledger parameters, four-term decomposition	Theory input

Table 3: Key references and their roles in the proposed calculation.

5 Proposed Collaboration Structure

Task	**HAL QCD**	**UIDT / P. Rietz**
Gauge configurations (existing)	✓	—
NBS wave function calculation	✓	—
$\lvert\psi_{cc}(0)\rvert^2_{\text{lat}}$ extraction & uncertainty	✓	—
UIDT parameter input (Δ^*, γ, α_s)	—	✓
80-digit chromomagnetic recalculation	—	✓
Residuum & claim update	—	✓
Joint publication (PRL / PRD letter)	✓(shared)	✓(shared)

Table 4: Division of responsibilities in the proposed collaboration.

Expected outcome: A joint short paper ($\leq$ 6 pages, PRL format) demonstrating or falsifying the UIDT prediction for $\Delta M(\Xi_{cc})$ at the level of the dominant chromomagnetic contribution, thereby upgrading the evidence category from B to A$^-$ or triggering a [TENSION ALERT] for reformulation.

6 Known Limitations of the UIDT Framework

Limitation 6.1 (Phenomenological γ). The kinetic vacuum parameter $\gamma = 16.339$ is derived from the ratio Δ^*/K_S and is not analytically proven from first principles. Evidence category: A$^-$ (phenomenological).

Limitation 6.2 (Electron Mass Residual). The UIDT prediction for the electron mass retains a residual correction not yet derived from first principles.

Limitation 6.3 (Cosmology Calibration). Cosmological parameters (w_0, E_T) are calibrated to DESI DR2 / Planck 2018 rather than derived independently. Evidence category: C.

7 Reproducibility and Open Science

Reproducibility Statement

- **Repository:** `github.com/Mass-Gap/UIDT-Framework-v3.9-Canonical`
- **Zenodo DOI:** `doi:10.5281/zenodo.19157809`
- **Precision:** `mp.dps = 80` (local, per UIDT Race Condition Lock); zero `float()` conversions throughout.
- **Reproduction command:** `pytest verification/tests/` (requires `mpmath`, `pytest`; no mocks; 80-digit residual checks).
- **Timestamp:** four-term decomposition committed to repository prior to LHCb publication.

8 Summary

Summary and Collaboration Request

The UIDT Framework predicts a cc-diquark NBS wave function at the origin of

$$\boxed{|\psi_{cc}(0)|^2_{\mathrm{UIDT}} = 0.4719\,\mathrm{fm}^{-3}}$$

derived from the LHCb-measured isospin splitting $\Delta M(\Xi_{cc}) = 1.43 \pm 0.76\,\mathrm{MeV}$ and the UIDT vacuum parameters $\Delta^* = 1.71003\ldots$ GeV, $\gamma = 16.339$. The extended HAL QCD method of your group [1] is the uniquely appropriate tool to test this prediction. We welcome the opportunity to discuss a collaboration at your convenience.

Contact: `uidt-framework@outlook.com`
Repository: `github.com/Mass-Gap/UIDT-Framework-v3.9-Canonical`

P. Rietz / UIDT Framework v3.9 - March 2026. All UIDT claims carry the evidence classification of the UIDT Evidence System (A–E). Limitations are stated in Section 6. Cosmological tensions such as the H_0 and S_8 tensions are not declared solved. Transparency has priority over narrative coherence.

References

[1] K.-W. Kelvin-Lee and N. Ishii, *Diquark mass and quark-diquark potential by lattice QCD using an extended HAL QCD method with a static quark*, arXiv:2601.10091 [hep-lat] (2026). arXiv:2601.10091

[2] K. Watanabe and N. Ishii, *Quark-diquark potential and diquark mass from Lattice QCD using the HAL QCD method*, Phys. Rev. D **105**, 074510 (2022). doi:10.1103/PhysRevD.105.074510

[3] M. Padmanath, R. G. Edwards, N. Mathur, and M. Peardon, *Spectroscopy of doubly-charmed baryons from lattice QCD*, Phys. Rev. D **91**, 094502 (2015). arXiv:1412.4782

[4] J.-Y. Yi, Z.-R. Liang, L. Liu, and D.-L. Yao, *Low-energy interactions between doubly charmed baryons and Goldstone bosons from lattice QCD*, arXiv:2511.12611 [hep-lat] (2025, rev. 2026). arXiv:2511.12611

[5] K.-W. Kelvin-Lee and N. Ishii, *Axial-vector diquark in Σ_c via extended HAL QCD method*, PoS(LATTICE2024)306 (2025). doi:10.22323/1.453.0306

[6] P. Rietz, *Unified Information-Density Theory (UIDT v3.9): Vacuum Information Density as the Fundamental Geometric Scalar*, Zenodo (2026). doi:10.5281/zenodo.17835200